The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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Reinforcement Learning

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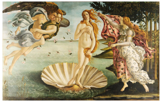
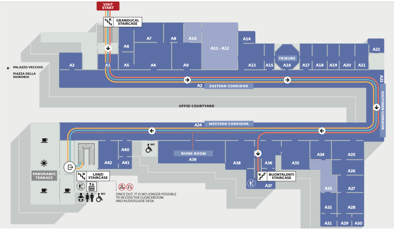
About 5,000 visitors a day move through ninety-eight rooms on two floors. These are the paintings nearly all of them come to see.

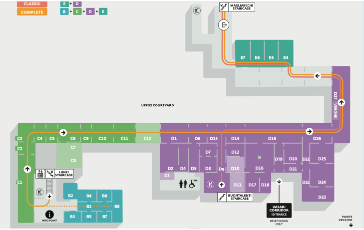




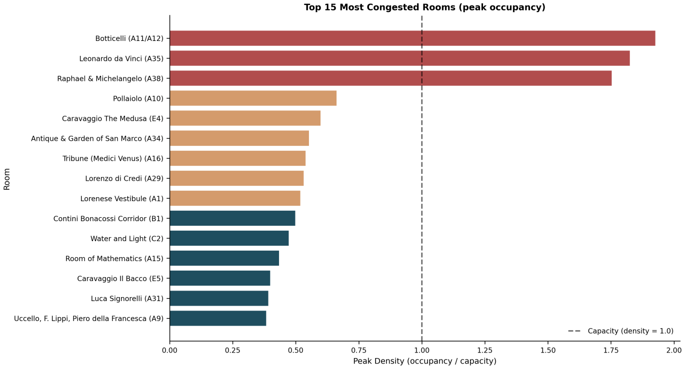

Second floor (the masterpiece rooms)



First floor (Caravaggio, free-flow)



The problem: where RL comes in



At the daily peak the three masterpiece rooms, Botticelli, Leonardo and Raphael, run at roughly twice capacity (red, past the dashed line) while the rest of the museum sits well below. The question: can reinforcement learning help us understand this crowding and simulate a calmer, better-run Uffizi?

The plan

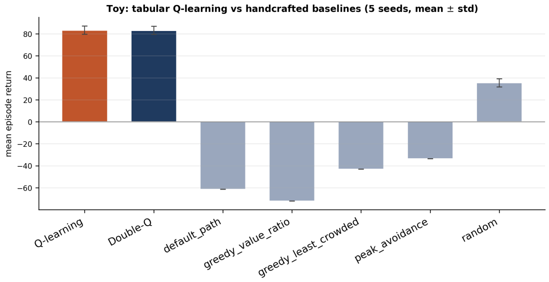
1. Recreate the Uffizi as a virtual model: every room, its size and doorways, plus a value for each (a Botticelli room is worth far more than a stairwell).
2. Generate the crowd, today's arrivals and mix, about 60% phone-first tourists, 30% ordinary visitors, 10% art lovers. Validate it against the density maps (does it look like the Uffizi today?), then read off today's revenue and average welfare.
3. Drop an RL agent on the museum as-is: the optimal visitor today.
4. Intervene: change hours, room access and pricing; re-simulate; keep only changes that hold or raise revenue while lifting welfare, because a revenue-losing redesign is never adopted.
5. Re-run the RL agents on the intervened museum: did the optimal visit improve for every kind of visitor?

A toy first: learning beats fixed rules

We prove it on a 12-room toy with an exact value table. Crowds wave up and down, unannounced.

Tabular Q-learning learns temporal arbitrage: enter in the lull around the crowd peak.

Across five seeds: Q and Double-Q tie (83.3 vs 83.1). A deterministic toy has no overestimation bias to remove, which is just what the theory says.



Learning beats every hand-written rule; the rules even go negative by mistiming the crowd.

The toy does not scale: deep RL and masking

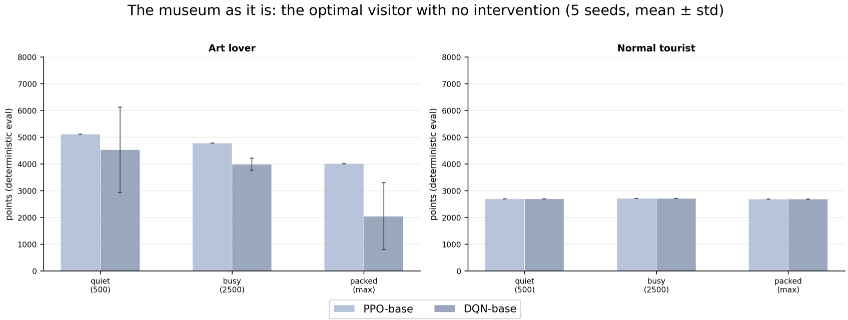
The real museum breaks the table: ninety-eight rooms, $\sim$100 moves per step, hundreds of state numbers. The same Q-table could never fill, so we change algorithm, to a neural net (MaskablePPO; PPO and DQN as controls).

Action masking: the rulebook

From any room only a few moves are legal, yet the net has $\sim$100 outputs. We set every illegal move to $-\infty$ before the softmax, so it gets exactly zero probability. So no gradient goes into the moves that cannot happen; the agent only ever picks among the legal ones.

$$\pi(a \mid s) = \frac{e^{z_a}\,\mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b}\,\mathbf{1}[b \in \mathcal{A}(s)]}.$$

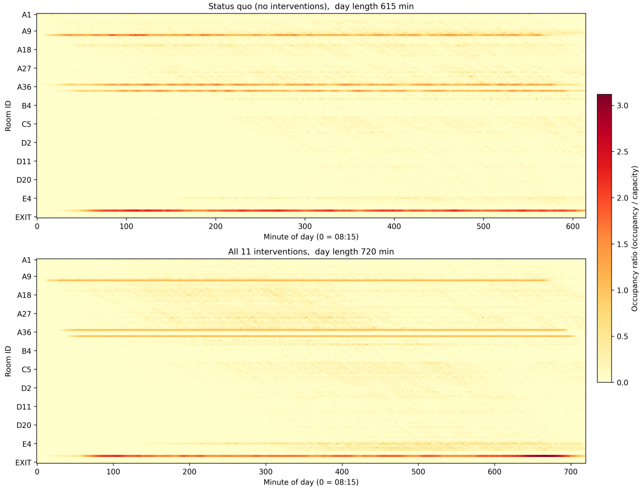
The museum as it is: the baseline



With the virtual museum calibrated to today (real hours, prices and surcharges), we drop the RL agents on the museum as it is, where the visitor just walks. This is the ceiling with no intervention: the art lover slips from ∼5,100 to ∼4,000 as the rooms fill; the tourist holds near ∼2,700. That is today's Uffizi, the baseline we compare everything against.

The interventions and the museum-wide payoff

Crowd density heatmap: status quo vs all 11 interventions (~5000 visitors). Shared colour scale.



Eleven Pigovian measures, each pricing the crowding a visitor causes:

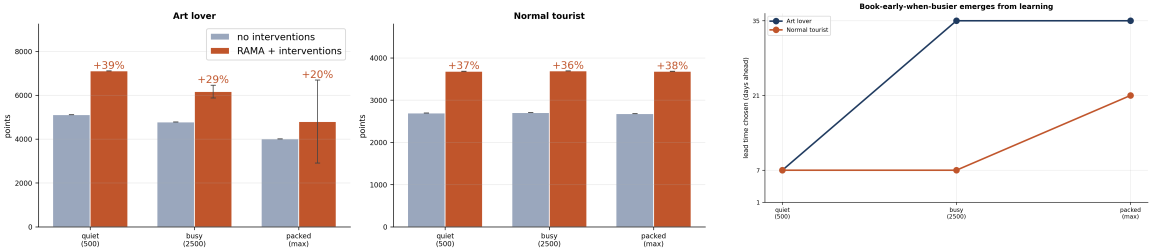
- Time: longer hours, quiet-hour discounts.
- Access: reservations (RAMA), enrich the side rooms.
- Price: peak pricing, group surcharge.

Top: as-is; bottom: all eleven, same scale. The masterpiece bands cool from red toward orange; the day spreads over more hours.
The director's test is met: +31% welfare, revenue held.

After the redesign: the visit becomes a booking problem

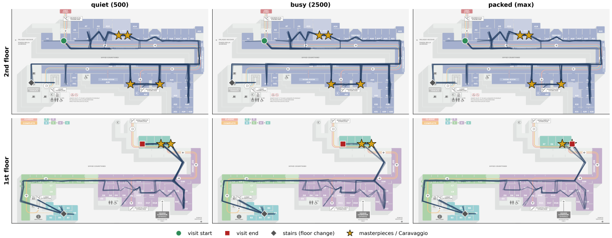
RAMA gates the masterpieces behind booked slots, so a visitor can no longer just walk in. The real decision flips from where to walk to when to book (lead time 1, 7, 21 or 35 days), hard because it is uncertain, irreversible and delayed.

RAMA booking: intervened vs matched baseline (5 seeds, mean ± std; 3/3 masterpieces secured)

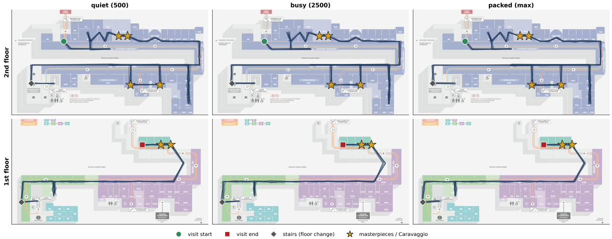


The ceiling rises at every crowd level. The art lover's baseline ∼5,100/4,000 climbs to ∼7,100/4,800 (+20 to +39%); the tourist's ∼2,700 climbs to ∼3,690 (+36 to +38%), all three masterpieces secured. We never coded book-early-when-busier (right); it came straight out of the reward.

Art lover walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



Normal tourist walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



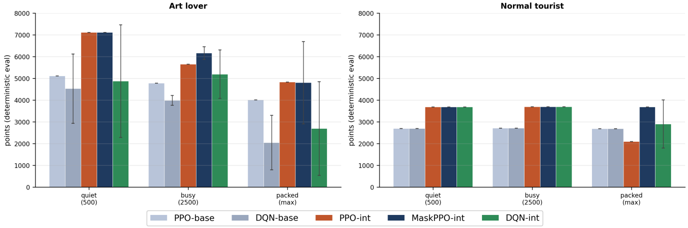
Top: the art lover. It works the whole museum: the trace covers the second floor end to end and fans across the first-floor blocks too.

Bottom: the normal tourist. It does not. After the masterpieces it skips the B block and hurries through C and D straight to Caravaggio, then leaves.

Same museum, two very different visits.

(Green = start, red = exit; stars mark the masterpieces and Caravaggio.)

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean ± std)



- Pale bars: the two no-booking baselines. Coloured: the three booking agents.
- MaskablePPO (dark blue) is best or tied in every cell and stable across seeds: the winner.
- Unmasked PPO (orange) collapses at the packed tourist (2093, below the 2680 baseline): it wastes effort on illegal moves. Masking only matters at the hard cells, but there it decides the result.
- DQN (green) is crowd-fragile and high-variance (the wide error bars): a max over noisy crowd values overestimates, the bias the deterministic toy did not have.

1: an unlearnable reward

S Symptom. The agent never left, squatting in one room until closing. Zeroing all reward for not exiting made it worse.

D Diagnosis. An all-or-nothing wipe is flat almost everywhere, so the gradient is zero: nothing points to the door.

F Fix. Use a dense egress signal: per-step pressure rising near closing and far from an exit, plus a bonus for leaving.

L Lesson. Even a correct reward is useless if it is flat everywhere. Gradient descent needs a slope to follow, so we built one in.

2: it could not see what it was deciding

S Symptom. The rule “stay until satisfied, then move” never stabilised; dwell times wobbled at random.

D Diagnosis. The right action depended on value already taken from the room, not in the state: a hidden variable the agent could not see.

F Fix. Add a per-room appreciation-progress value to the state, so “seen enough?” becomes something the policy can read.

L Lesson. Whatever the best move depends on has to live in the state. We had left it out, so no policy could settle.

3: reward shaping that backfired

S Symptom. Well-meant terms backfired: a “boredom” penalty caused looping, a harsh crowd penalty made a fake art lover.

D Diagnosis. Each reward term we added gave the agent a new way to cheat: looping beat exiting, a crowded Botticelli scored worse than just skipping it.

F Fix. Minimal shaping: a gentle crowd discount so a crowded masterpiece still beats skipping, no boredom penalty, satiation alone stops lingering.

L Lesson. The agent games every term the laziest way it can. We stopped adding shaping until we had checked what each term really paid for.

4: the do-nothing trap and the book-early trick

S Symptom. Given a free “decline” button, both profiles declined all three masterpieces; book-early stayed only half learned.

D Diagnosis. Declining is safe now, booking pays only later: a do-nothing optimum; its reward lands hundreds of steps away (credit assignment).

F Fix. Make “always books” a type trait with no escape hatch; add an immediate off-pace penalty proxying the distant cost.

L Lesson. A small penalty now, standing in for a cost hundreds of steps away, was what finally got it booking. Nothing else we tried helped as much.

Conclusion

We rebuilt the Uffizi, matched it to the real museum, then asked two questions: how to redesign the building and how a single visitor should play it.

- The redesign: eleven Pigovian measures lift welfare $+31\%$ with revenue held.
- The visitor: the optimal agent (MaskablePPO) books early and beats its as-is ceiling at every crowd level.
- The lesson: choosing what the agent decides was far harder than the algorithm itself; action masking did more for us than any amount of tuning.

Thank you.

Questions?

Appendix

Backup slides for questions

Backup: the three MDPs at a glance

Setting	State	Action	Reward	γ
Toy (Q, Double-Q)	5-tuple table	stay / move	toy per-step	0.99
As-is (PPO, DQN)	8 numbers	stay / move on	walk, no booking	0.995/0.997
RAMA (MaskablePPO, PPO, DQN)	25 numbers	lead / book / pace masked for MPPO	booking per-step	0.999

The same five elements in every setting; what changes is the problem the agent is attached to, plus how the state is stored and the actions screened. PPO and DQN appear twice, once as a no-booking walk (the baselines) and once as booking agents; MaskablePPO only on RAMA.

Backup: the as-is state, eight numbers

The visitor's local view on the planned walk (natural units; the network sees each scaled to about $[0, 1]$):

1. Room-value drain: minutes absorbed vs the room's worth ($\sim$3 min per importance point).
2. Room value: importance, 1 to 10.
3. Density: how full the room is, percent of capacity.
4. On-target flag: are you in the next planned must-see?
5. Plan completed: share of must-sees finished.
6. Time of day: minute of the 615-minute day.
7. Egress slack: spare minutes to reach an exit before closing.
8. Distance to next room: doorways to the next must-see.

At the door `A1`: (0, 0.22, 0.30, 0, 0, 0, 0.99, 0.10).

Backup: the RAMA state, twenty-five numbers

A two-phase episode (a booking screen, then the walk), grouped:

- Booking position (3): phase flag, which of 4 decisions, lead chosen (days).
- Booking screen (4): slots free at lead 1/7/21/35 days (out of 8).
- Time of day (1): minute of the 720-minute day.
- Windows held (3): booked clock-times for Botticelli, Leonardo, Raphael.
- Check-ins (3): which masterpieces collected.
- Access read (4): is-masterpiece, unbooked, open-now, minutes-to-open.
- Pace prior (3): when it expects to reach each masterpiece (frozen).
- Room tail (4): drain, value, density, egress, as in the as-is state.

Opening screen: $(1, 0, 1;\ 0.12, 0.38, 0.62, 1;\ 0;\ \ldots)$; the booking screen is the crux.

Actions per setting

- Toy / as-is: `Discrete(2)`, stay or move on.
- RAMA: `Discrete(4)`. Booking phase: a lead $\in \{1, 7, 21, 35\}$ or book/decline. Walk phase: stay or move on.

Masking (MaskablePPO). Set the logits of illegal moves to $-\infty$ before the softmax, so they get exactly zero probability:

$$\pi(a \mid s) = \frac{e^{z_a}\,\mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b}\,\mathbf{1}[b \in \mathcal{A}(s)]}.$$

Backup: the reward functions

Toy, per step. $\dfrac{\text{importance} \times \text{novelty}}{1 + \alpha_A \,\text{density}^2} - \text{step cost}$, with $\alpha_A = 6$, step 0.5, exit bonus $+15$, no-exit penalty -50 (hand-tuned shaping; the relative sizes are what matter).

Full museum, per minute. importance $\times$ crowd discount $\times$ novelty (satiation), with a gentle crowd $\alpha = 0.5$, plus closing-time pressure, a completion bonus and a small step cost.

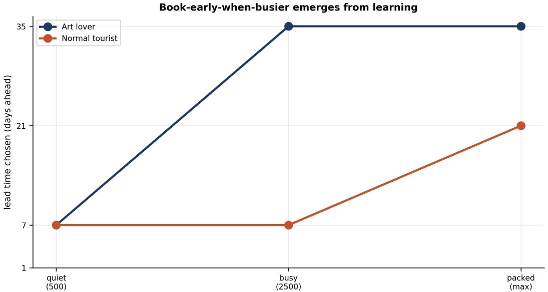
Booking terms (RAMA only). an immediate off-pace penalty at commit time (a local proxy for the distant cost of a badly-timed slot) plus a no-show penalty for a masterpiece booked but missed.

Backup: the booking phase and the fill curve

Two-phase episode

- Booking (no time passes): pick a lead time (1/7/21/35 days), then book each masterpiece.
- Windows committed near the expected arrival; then the minute-by-minute walk.

Fill curve (busy day). Slots free out of 8: 1 a day before, 3 a week ahead, 5 three weeks ahead, 8 a month ahead. Book late, miss the masterpiece.

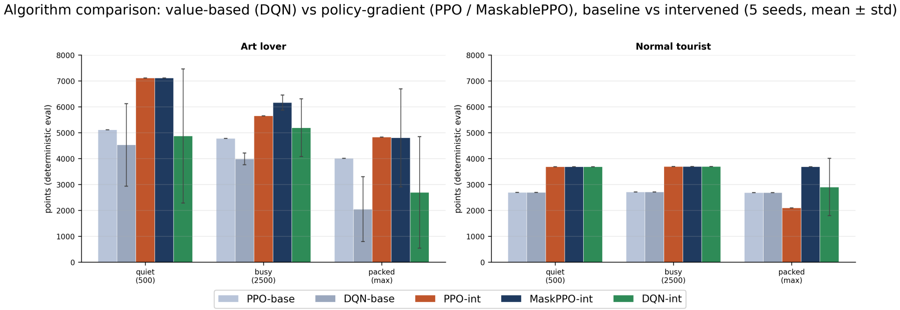


Backup: full booking results, five seeds (mean $\pm$ std)

Profile	Crowd	Baseline	Intervened	Gain	Check-ins
Art	500	5112	7113 ± 0	$+39\%$	$3.0/3$
Art	2500	4780	6167 ± 291	$+29\%$	$3.0/3$
Art	5000	4010	4800 ± 1891	$+20\%$	$2.6/3$
Tourist	500	2694	3686 ± 0	$+37\%$	$3.0/3$
Tourist	2500	2708	3695 ± 0	$+36\%$	$3.0/3$
Tourist	5000	2680	3685 ± 0	$+38\%$	$3.0/3$

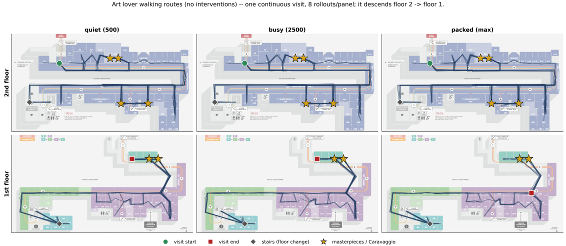
Baseline = best no-booking walk (ppo_base); intervened = MaskablePPO on RAMA. Five training seeds $\{1, 3, 7, 42, 202606\}$, each evaluated on six fixed common-random-number crowds. The art lover at max crowd is bimodal: four seeds reach $\sim$+41%, one collapses, hence the wide band and the 2.6/3 check-ins.

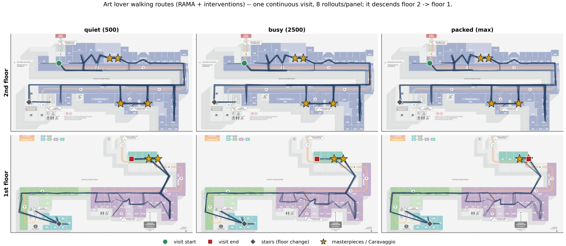
Backup: which algorithm and where it breaks



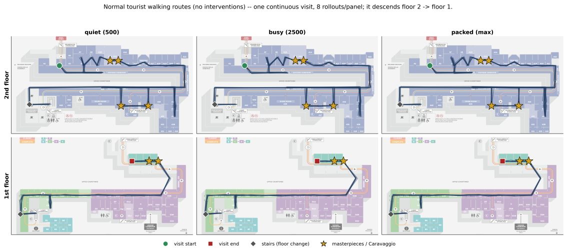
- MaskablePPO: best or tied in every cell, near-zero seed variance.
- Unmasked PPO: collapses at the packed tourist to 2093, below the 2680 no-booking baseline, while MaskablePPO holds 3685.
- DQN: crowd-fragile and high-variance, its art baseline falls 4527 $\rightarrow$ 2044 as the crowd grows (a max over noisy values overestimates).

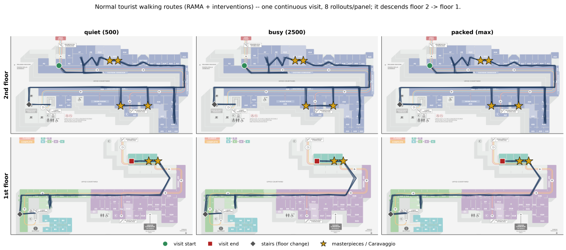
Backup: the learned walk, art lover





Left: as-is. Right: under RAMA. Eight rollouts per panel across the three crowd levels; the masterpieces (gold stars) are taken in their booked windows, then a descent to Caravaggio and out.

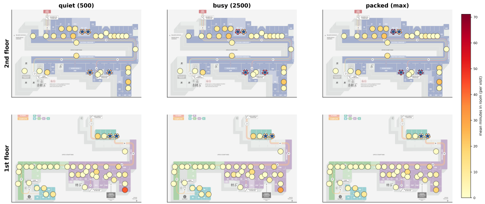




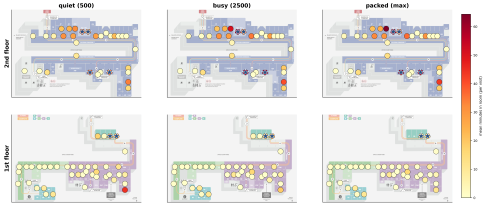
Left: as-is. Right: under RAMA. The tourist books all three masterpieces and stays rock-stable across seeds at every crowd level.

Backup: per-profile crowd density

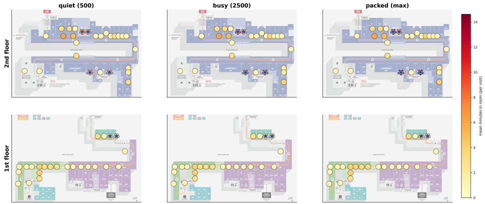
Art lover -- where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



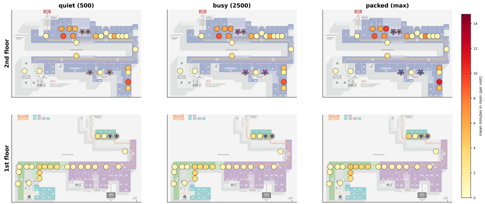
Art lover -- where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Normal tourist -- where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

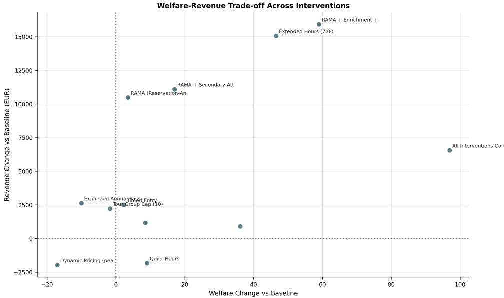


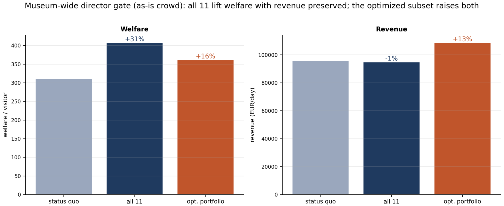
Normal tourist -- where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Left column as-is, right column under RAMA; art lover on top, tourist below. The booked windows spread each profile's masterpiece visits across the day.

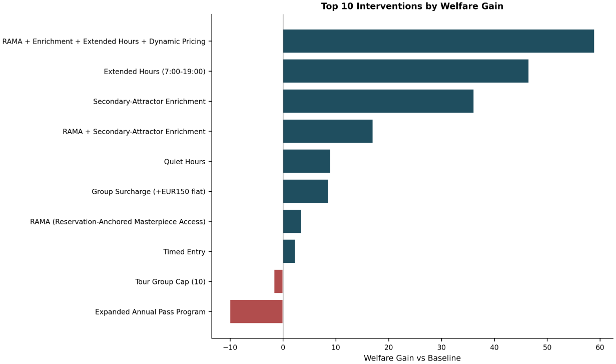
Backup: the interventions and the director's gate





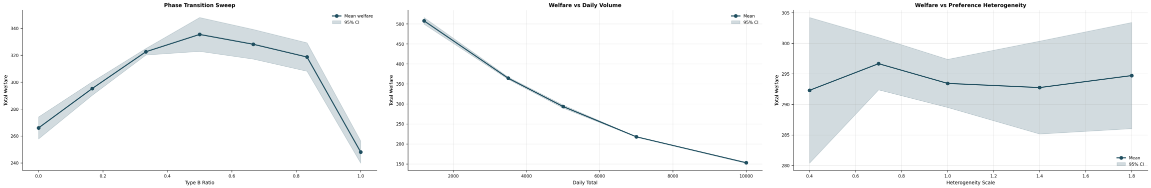
The redesign is adopted only if welfare rises and revenue does not fall (the director's gate). The chosen portfolio of eleven Pigovian measures lifts welfare +31% with revenue preserved.

Backup: ranking the candidate interventions by welfare gain



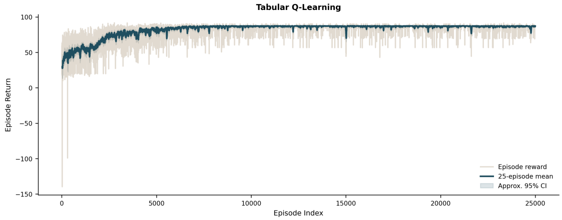
Each candidate measure scored on its own against the as-is baseline. Extended opening hours and secondary-attractor enrichment are the strongest single levers; the full combined bundle leads. The tour-group cap and the expanded annual pass lower welfare on their own; the adopted portfolio keeps the welfare-positive measures.

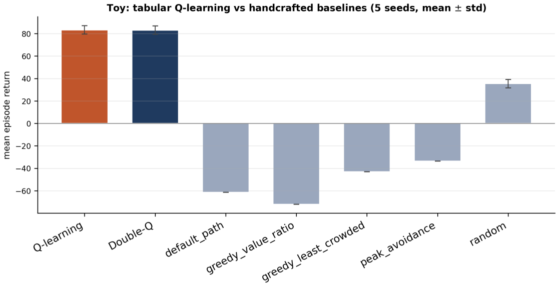
Backup: robustness sweeps



We vary the Type-B (Instagram) share, the total daily volume and the crowd heterogeneity. The welfare gain from the redesign holds across the ranges we tried.

Backup: tabular toy internals





Left: the learning curve, return climbing over episodes. Right: learning beats every hand-written rule. Across five seeds Q (83.3 ± 3.8) and Double-Q (83.1 ± 3.7) tie, since a deterministic toy has no overestimation bias for Double-Q to remove.

Backup: training and evaluation setup

Deep RL (Stable-Baselines3). `MlpPolicy` $[128, 128]$, Adam LR 3×10^{-4}, 2048 steps, batch 256, 10 epochs, GAE $\lambda = 0.98$, clip 0.2, entropy 0.01.

- As-is baselines (PPO): 150k steps, γ 0.995/0.997.
- RAMA arms: 100k steps, $\gamma = 0.999$.

Toy. tabular Q / Double-Q, ε-greedy, $\gamma = 0.99$, 25k episodes.

Seeds and evaluation

- 5 training seeds $\{1, 3, 7, 42, 202606\}$.
- Eval on 6 fixed crowds (900000 to 900005), deterministic, common random numbers.
- Crowd levels 500 / 2500 / 5000 visitors a day.

Backup: limitations and where we would go next

One visitor, not a crowd. What we optimise is a single sophisticated visitor's best response to a fixed crowd. It is not an equilibrium: if every visitor booked this way the good slots would fill faster and the fill curve itself would move. Closing that loop, a mean-field game over the whole population, is the real next step.

Fixes we did not run. Our toy isolates the overestimation that makes DQN crowd-fragile, yet we never tried the obvious cure, Double-DQN, on the full task. Two experiments would sharpen the story: a fixed-heuristic booker ("always reserve a month out") as a tougher baseline than no-booking, plus an ablation that switches the off-pace penalty off to confirm that booking early is learned rather than coaxed.

What is baked in. The connoisseur always books (a trait, not a trick to win); lead times are four discrete values; the fill curve is calibrated rather than fitted to real booking data; the policy is a plain feedforward net carrying appreciation-progress, with $\gamma = 0.999$. Welfare, finally, is the simulator's own visitor-utility, itself a modelling choice rather than a settled social objective.